Wildlife Rescue Chain-of-Custody

Blockchain-Based Wildlife Custody and Verification System

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1. Problem Statement

Wildlife rescue involves multiple parties such as rescue teams, veterinary hospitals, rehabilitation centres, forest departments, and conservation organizations. When an animal moves between these entities, maintaining a trustworthy record of custody, treatment, rehabilitation, and release becomes difficult.

Conventional records may be distributed across organizations and can be difficult to verify. The proposed system uses blockchain and Solidity smart contracts to create a tamper-resistant chain of custody, enforce authorized transfers, track important lifecycle events, and provide verifiable evidence of an animal's rescue and rehabilitation history.

2. Proposed System

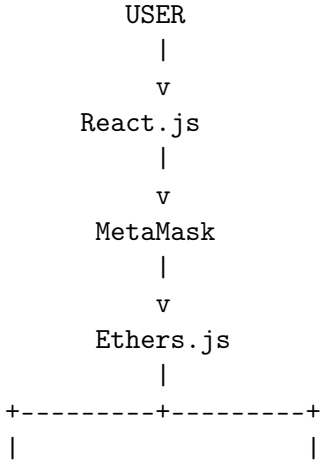
Wildlife Rescue Chain-of-Custody assigns every rescued animal a unique digital identity. Authorized organizations interact with the system through a web interface and blockchain wallet.

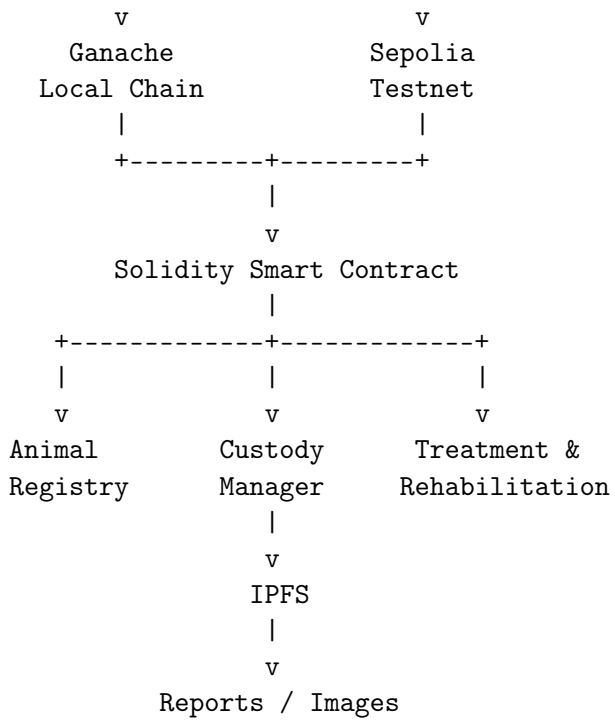
The smart contract maintains the animal's current custodian and lifecycle state. Custody transfers require authorization and acceptance by the receiving organization. Treatment and rehabilitation milestones can also be recorded before the animal is approved for release.

Sensitive files such as photographs and medical reports are stored off-chain using IPFS. Their cryptographic hashes are stored on the blockchain, allowing the original files to be verified later without storing large files directly on-chain.

3. System Architecture

The proposed architecture combines a web application, Ethereum-compatible blockchain, Solidity smart contracts, and decentralized storage.





Ganache is used as the local Ethereum blockchain during development and testing, while **Sepolia** can be used for testnet deployment. MetaMask provides user authentication and transaction signing, while Ethers.js connects the frontend to the smart contract.

4. Wildlife Rescue Lifecycle

The system models the animal's lifecycle using predefined smart contract states:

```
RESCUED -> TRIAGED -> UNDER_TREATMENT
        -> RECOVERING -> REHABILITATION
        -> RELEASE_READY -> RELEASED
```

Invalid transitions are rejected by the smart contract. For example, an animal cannot directly move from `RESCUED` to `RELEASED` without satisfying the required conditions.

The general process is:

```
   |
   v
 Authorized Release
```

5. Smart Contract Functionality

The core blockchain functionality can be implemented through the following modules.

- **Animal Registry:** Registers an animal and stores its unique identifier, species, status, and current custodian.
- **Custody Manager:** Handles custody transfer requests and verifies that both the sender and recipient are authorized organizations.
- **Treatment Manager:** Records treatment, recovery, and rehabilitation milestones.
- **Authorization Manager:** Maintains the list of organizations and authorities permitted to perform specific operations.

A custody transfer follows a two-party verification model:

```
Current Custodian
       |
       v
Transfer Request
       |
       v
Smart Contract
       |
       +-- Verify Sender
       +-- Verify Recipient
       |
       v
Receiving Organization
       |
       v
Accept Transfer
       |
       v
Custody Updated
```

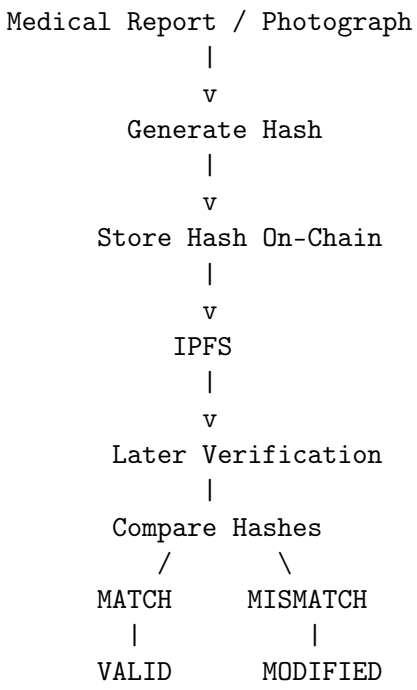
This prevents an organization from claiming custody without the transfer being authorized and accepted.

6. Blockchain and Off-Chain Data

The blockchain stores important metadata such as:

- Unique animal ID and species
- Current and previous custodian
- Custody transfer history
- Lifecycle status
- Treatment and rehabilitation milestones
- Release authorization
- Event timestamps
- Supporting document hashes

Large or sensitive data should remain off-chain. Medical reports, photographs, and other evidence can be stored in IPFS, while their hashes are recorded on-chain.



7. Key Features and Novelty

The project focuses on the following features:

1. **Immutable Chain of Custody:** Every authorized transfer creates a permanent blockchain record.
2. **Smart Contract Authorization:** Only registered organizations can perform permitted custody operations.
3. **Controlled Lifecycle:** Smart contracts prevent invalid animal status transitions.
4. **Two-Party Custody Acceptance:** A receiving organization must accept custody before the transfer becomes final.
5. **Treatment Verification:** Treatment and rehabilitation milestones can be recorded and verified before release.
6. **Evidence Integrity:** Blockchain-stored hashes allow off-chain documents to be checked for modification.
7. **Complete Animal History:** Rescue, custody, treatment, rehabilitation, and release events form a chronological and verifiable record.

The key novelty is applying blockchain's immutability and smart-contract-based authorization to the **chain of custody of rescued wildlife**, rather than using blockchain merely as a generic record-keeping database.

8. Technology Stack

Component	Technology
Frontend	React.js, Tailwind CSS
Smart Contracts	Solidity
Blockchain Development	Hardhat
Local Blockchain	Ganache
Test Network	Ethereum Sepolia
Wallet	MetaMask
Blockchain Interface	Ethers.js
Off-chain Storage	IPFS

9. Expected Outcome

The final prototype should demonstrate:

- Registration of rescued animals
- Authorized custody transfers
- Blockchain-based lifecycle tracking
- Treatment and rehabilitation milestone recording
- Release authorization
- Verification of off-chain evidence using blockchain hashes
- A complete tamper-resistant custody history

The project demonstrates a practical application of Solidity and blockchain technology to a real-world conservation problem involving trust, accountability, traceability, and verification.

10. Conclusion

Wildlife Rescue Chain-of-Custody provides a decentralized approach to tracking rescued animals from rescue to release. Solidity smart contracts enforce authorization and lifecycle rules, while blockchain records provide an immutable history of important events.

By combining on-chain records with IPFS-based evidence storage, the system can provide both transparency and practical data management. The project therefore demonstrates how blockchain can be applied beyond financial systems to support wildlife conservation and accountable rescue operations.